
NESTED-SPACE COSMOLOGY

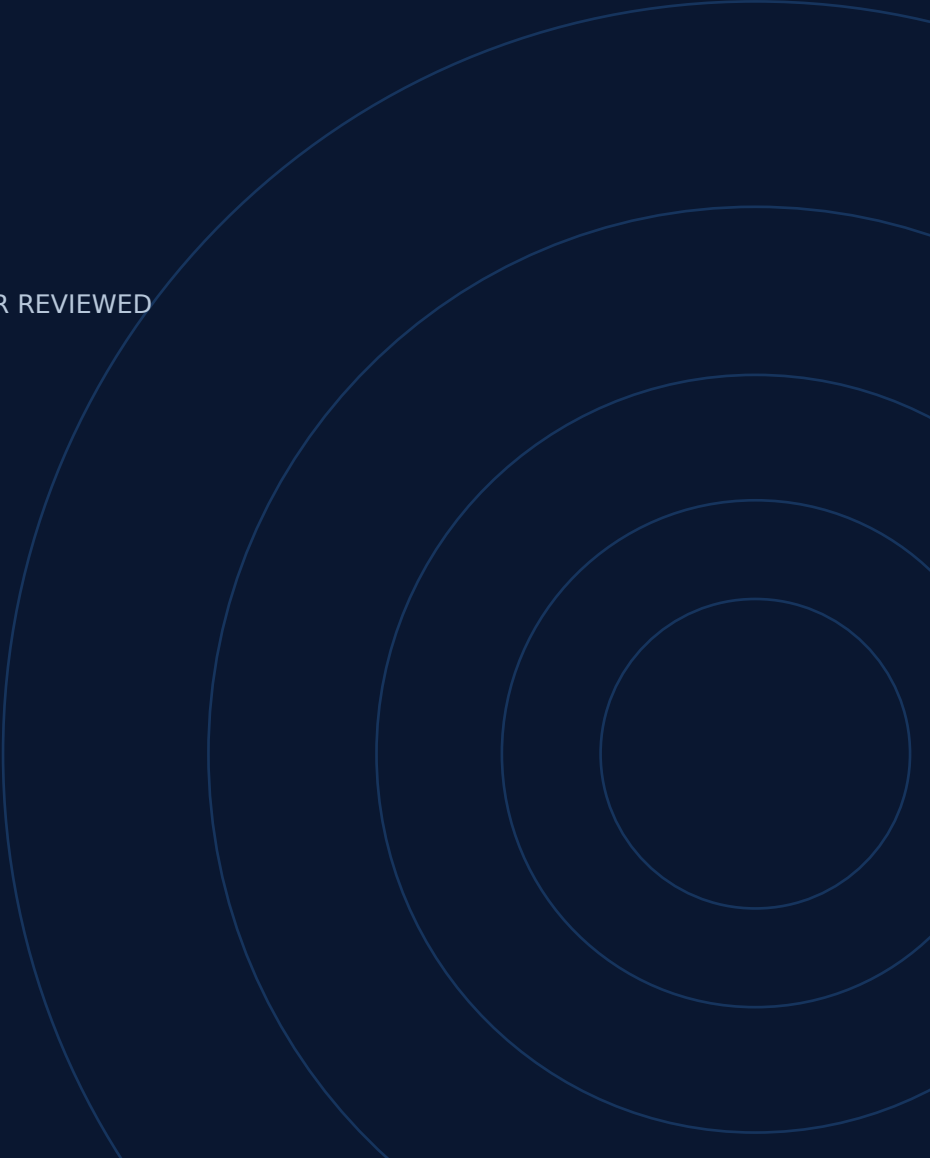
One Recursive Spectral Gradient–Boundary Equation
Across Scales

*A constructive one-equation framework and its current reproducible
evidence*

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Abstract

Nested-Space Cosmology proposes that local matter, inherited constants, unresolved gravitational response, and a finite parent-to-child transition arise from one recursive operator and one set of laws. The organizing condition is $T_{\Theta}^* D_{\Theta} = D_{\Theta}$. This paper separates that physical hypothesis from exact operator identities, computed development models, and observations still to be predicted.

The verified chain now contains evaluated full-spinor finite throat maps, normalized recursion, a gapless isolated radial operator, smooth periodic geometry with a band gap, causal Dirac propagation on a horizon benchmark, covariant vacuum-source controls, and quantum stress differences on a varying neck. A prescribed radius deformation produces Dirac pairs whose energy is accounted for by geometric work. A scalar sheet coupling admits an invariant chiral sector equal to an ordinary massive Dirac Hamiltonian. The same covariant frequency operator now supplies an explicit ultraviolet-subtracted determinant source, an exact comparison of normalization profiles, and a normalized finite state functional whose smeared noise equals the pair-production weight. These results connect geometry, matter-like excitations and response; they do not determine a scheme-independent completed stress, a physical mass spectrum, or a self-consistent cosmology.

The next matching equation is the finite completion, compensator, physical state and link of that specified determinant source, together with the actual boundary interaction. The calculations distinguish stored vacuum stress from energy transport, identify finite action terms requiring ultraviolet matching, and provide precise tests of the physical chiral sector. The 81-record collection preserves the original 58 historical records and adds 23 scoped follow-ups. This working formulation provides an explicit route to physical scale, matter and cosmological predictions; those predictions remain to be obtained from the coupled solution.

1. One hypothesis and a continuous physical test

Postulate. Spaces form inside spaces through collapse, localization and renewed expansion. A room can change its configuration and locally measured scales while inheriting the same dimensionless law. The central requirement is

$$\mathcal{T}_{\Theta}^* D_{\Theta} = D_{\Theta}.$$

Its block form has room operators on the diagonal and parent/child boundary maps off diagonal. The proposed invariant partition and present spectral prescription are

$$Z_{\Lambda}^{\text{one}}(D_{\Theta}) = \int \mathcal{D}\phi Z_{\Lambda}(e^{-\phi/2} D_{\Theta} e^{-\phi/2})$$

$$S_{\text{one}} = \text{Tr} e^{-D_{\Theta}^2/\Lambda^2} + \langle \Psi, D_{\Theta} \Psi \rangle.$$

The displayed functional is a formulation to complete, not an already normalized quantum-gravity path integral. The regulator, normalization scale, compensator, measure, determinant phase, zero modes and physical domain must be derived consistently. Induced geometric terms must be counted once. Adding a separate gravitational weight to rescue a stationary root would change the adopted construction.

The same action must generate the source of its geometry, the physical particle sectors and their interactions, and the observable response. The operational chain is:

```
common action + state + recursive boundary domain
-> boundary response and absolute stress
-> coupled geometry and state evolution
-> matter, energy transfer and physical modes
-> expansion, clustering and lensing
-> independent observational tests
```

Douglas Ek is the accountable author and retains scientific and publication responsibility. ChatGPT/OpenAI Codex assisted with formulation, derivation, software, computation and presentation; Grok supplied bounded independent checks. The author identifies the principal collaborating sessions as GPT-5.6 Sol and GPT-6 Astra. These attribution labels do not certify a calculation: each result is supported by its committed equations, reproducer, source hashes and stated scope.

2. Six explanations the programme aims to establish

Each row is a research target. The third column records the calculation that would turn its interpretation into a physical result.

Target	Strongest present connection	Required physical closure
Constants and thermal limits	Characteristic/current calculations; scale, covariant-measure and normalization identities	Common causal cone, derived gravitational coupling and thermal state
Finite black-hole continuation	Regular benchmark geometry; horizon-penetrating Dirac transport	Self-sourced collapse, physical stability and complete continuation
Dark matter and dark energy	Boundary self-energy, ultraviolet-subtracted determinant source and energy-transfer identities	Joint background stress and perturbation predictions
Particle, wave and measurement	Field excitations, imported collective-coordinate reduction and a finite Gaussian state functional	Derived interactions, detector records and quantum statistics
Mass, chirality and antimatter	Exact invariant sheet/chirality sector and charge algebra	Actual boundary coupling, physical sector selection and gauge representations
Probability and recursive infinity	Tail endpoint criteria, clock-map controls and a specified determinant remainder	Consistent recursive state measure, causal duration and observable limits

2.1 Constants, resolution and absolute zero

Open prediction. The causal cone must follow from the physical principal symbol and agree across the relevant propagating species. A numerical value of a dimensional speed also depends on clock and ruler conventions. Derive the gravitational response before identifying

$$l_p = \sqrt{\hbar G/c^3}$$

with a scale of the model. A spectral cutoff does not by itself prove a shortest measurable distance; that requires a localized-probe/backreaction experiment. The historical constant dictionary names intended outputs but does not derive nature's values ([constant dictionary](#)).

Open prediction. Zero temperature is a limit of the physical many-body thermal state, not the bottom of the signed first-quantized Dirac spectrum. A suitable equilibrium construction must be independent of the arbitrary energy origin, handle ground-state degeneracy and zero modes, and distinguish residual spin or vacuum correlations from thermal excitation. The historical phrase equating absolute zero with a spectral infimum is therefore a proposed dictionary entry requiring this additional derivation.

2.2 Black holes and historical horizons

Imported result. Regular black-universe solutions already combine a trapped region, positive minimum areal radius and expanding interior in one geometry [6]. The project uses that solution as a development benchmark. Its existence is not proof that our universe is such an interior or that the same geometry is sourced by this operator.

Repository derivation. The actual horizon-penetrating radial Hamiltonian includes the shift and spin connection. Conserved Dirac norm transfers in the child direction in the declared Cauchy problem ([tetrad](#), [transport](#)). The trapped finite boundary has an outflow structure; it cannot be replaced by an arbitrary reflecting wall.

Open prediction. Complete physical continuation requires a dynamically sourced solution, constraints, finite tidal response and extension of relevant finite-affine endpoints. Infinite coordinate time is not affine completeness ([clock and horizon audit](#)). High-energy Euclidean spectral form factors [13] are not by themselves a Lorentzian theorem that bosonic signals stop at a physical wall.

2.3 Dark response from adjacent rooms

Postulate. Degrees of freedom outside a locally resolved room can act through its boundary response.

Repository derivation. Eliminating those degrees of freedom gives a Schur self-energy. This establishes an exact response mechanism, not its identification with the observed dark-matter and dark-energy components.

An isotropic background source and a clustering/lensing response are different projections of the same required effective stress. Both must be calculated. A finite vacuum cutoff does not solve the cosmological-constant problem without a predicted renormalized curvature and physical scale. Nor does a nonzero boundary map establish a continuing energy supply.

2.4 Localized excitations, waves and measurement

Imported result. Individual detections build interference patterns [1]. QFT describes particles as field excitations; solitons and collective coordinates supply additional useful construction patterns [4]. The existing [particle/wave record](#) is an imported collective-coordinate reduction. It does not solve an electron or detector.

Open prediction. Couple an identified excitation to a detector through the derived interaction and evolve the whole energy/charge ledger. Interference, localized records, outcome probabilities, Bell correlations and no-signalling must follow from that model. Decoherence or mechanical resonance alone does not derive the Born rule or select an individual outcome. Bell and PBR results constrain admissible accounts [2,3]; unresolved room labels alone do not establish one.

2.5 Chiral mass and charge-conjugate states

The scalar sheet benchmark in Section 3 provides an exact version of the proposed left/right relation. Its coupling has not yet been derived from the throat. Chirality, radial component, angular sector, sheet label and charge are distinct operations. Antimatter requires the conjugate gauge representation, not merely an opposite sheet or a reversed spatial gradient.

With the stated gamma conventions, distinguish the charge-conjugation matrix $C=i\gamma^2\gamma^0$ in $\psi^c=C\bar{\psi}^T$ from the antilinear column-spinor map $\psi^c=i\gamma^2\psi^*$. The quantum symmetry operation and a c-number conjugation map are not interchangeable definitions. Physical photons require a spin-one gauge sector. The produced Dirac pairs do not establish baryonic dust or the observed matter excess.

2.6 Probability, infinite generation and existence

Interpretive hypothesis. Infinity describes ongoing generation rather than one static container.

Repository derivation. On a particular unweighted spatial chain, the endpoint is uniquely determined for $\Omega q \leq 1$, while $\Omega q > 1$ requires data at infinite depth ([tail result](#)). This is an endpoint criterion, not a universal strict-contraction theorem; critical recursions can converge algebraically.

A nonterminating decimal is not evidence of nested geometry. Infinite room count is not proof of infinite mapped proper time. A physical claim of eternal generation requires the state measure, clock map, causal continuation and limiting observable to be defined. Energy conservation and a finite cutoff alone do not prove that child formation is inevitable or that existence has probability one.

3. A precise shared mass and self-energy benchmark

Repository derivation. In sheet-first Weyl order, consider

$$H_8 = I_2 \otimes \boldsymbol{\alpha} \cdot \mathbf{p} + \Phi \tau_1 \otimes \beta$$

$$\Pi_- = (I_8 - \tau_3 \otimes \gamma^5)/2.$$

The projector selects parent-left plus child-right. Exact calculation gives

$$[H_8, \Pi_-] = 0$$

$$W^\dagger H_8 W = \boldsymbol{\alpha} \cdot \mathbf{p} + \Phi \beta$$

where W embeds those four components. Thus one invariant sector is an ordinary massive Dirac Hamiltonian, with

$$E^2 = |\mathbf{p}|^2 + \Phi^2$$

$$\Sigma_p(E) = \Phi^2(E + \boldsymbol{\alpha} \cdot \mathbf{p})^{-1}.$$

The complementary sector is equally invariant and spectrally equivalent. Physical selection must follow from the common action or domain, not preference. A spinor-identity sheet link instead has energies $\pm p \pm \Phi$ and is gapless at $p = \Phi$. A Hermitian pseudoscalar link has the same free gap and self-energy as the scalar benchmark. The spectrum therefore does not determine the complete interaction. The [observable bridge](#) checks these distinctions and the corresponding charge algebra.

Both chiral components carry the same Dirac charge. A massive positive-energy eigenstate has a constant chirality expectation. The familiar left-to-right oscillation belongs to a specified initially chiral state containing both energy signs. In the full paired angular continuum problem, the massless radial geometry and its radius pulse preserve physical chirality. Their radial gap is consequently not already a four-dimensional chirality-breaking rest mass.

The next test is the actual boundary map's Clifford structure, orientation and invariant domain, followed by a controlled local limit and physical pole. These calculations are kinematic connections within this project; ordinary Dirac algebra is prior physics.

Repository derivation: evaluated boundary channels. The full paired-angular calculation now evaluates an energy-dependent four-component spatial link in an explicit current-conserving domain. Direct joined, transfer and Schur calculations agree below $2e-15$ relative; an independent continuum integration gives 0.00947%–0.03645% errors at the reported resolution. The link preserves physical chirality. Its action-form Clifford decomposition has vector and axial content; scalar and pseudoscalar terms vanish to numerical precision ([chiral boundary](#)).

This result specifies the actual transport structure of that massless spatial realization. The proposed chiral-sector mass reduction requires an additional interaction or physical domain mechanism derived from the common action. A passive change of the child spin frame does not supply it: chirality must be transported with the frame. The numerical map is a finite spatial point-port response, with its units and regulator stated explicitly; it is not the full PG retarded map or a local mass term.

4. Geometry, vacuum stress and an explicit energy ledger

Repository derivation. The isolated radial throat is gapless. Periodic confinement can create a band gap, and a smooth periodic profile removes the earlier geometric seams:

$$r(x) = \sqrt{a^2 + [\sin(kx)/k]^2}$$

$$k = \pi/(2R).$$

At $a=1$ and $R=2,4,8$, its continuum first band edges are approximately 0.76423204, 0.51425546 and 0.28454085 ([smooth geometry](#)). R remains an input. These are spatial spectral values, not identified particle masses.

The same profile requires a negative effective Einstein null source,

$$8\pi G(\rho + p_x) = -2r''/r.$$

The [covariant vacuum controls](#) establish negative axial-null vacuum response on a specified compact constant cylinder. The [varying-neck calculation](#) differentiates all metric functions and computes a periodic-minus-antiperiodic stress difference. Its local finite counterterms cancel in that difference; its absolute stress is not thereby fixed. Choosing a spin structure because its stress has the desired sign would not derive it.

A finite term proportional to the squared Weyl curvature leaves the anomaly unchanged but changes the varying-neck shape equation. The current flat common-scale integral also diverges in its finite realization. The ultraviolet functional, complete compensator and state must resolve these questions before a stationary root represents self-sourcing.

Repository derivation: finite metric response. The general-metric calculation now isolates four independent bulk channels in the local basis $M^4, M^2R, C^2, R^2, E_4, \square R$. On the closed smooth cell, the Euler and divergence terms have exact zero bulk variation. Both $R=2$ and $R=4$ profiles distinguish the remaining four coefficients; the constant-cylinder control has rank two. Independent lapse, radial metric and radius variations agree with direct energy changes ([finite terms](#)).

For the declared finite-action sign convention, the exact neck null response is

$$\Delta(\rho + p_x)_{\text{fin}} = \frac{4M^2}{a^2}C_R + \frac{32(1+a^2k^2)}{3a^4}C_C - \frac{16(1+4a^2k^2)}{a^4}C_{R^2}.$$

This gives a concrete ultraviolet matching problem: determine these coefficients from the same functional, then solve the independent metric equations. They have not been fitted to make the neck stationary. The result concerns a closed torsionless metric sector through four derivatives; boundary contributions and other fields require their corresponding terms.

Repository derivation: covariant determinant source. The static Euclidean operator on the same smooth cell is

$$D_\omega = \beta\omega/N + i\beta H_0(q, r).$$

Physical four-dimensional chirality anticommutes with D_ω . Its square retains the lapse gradient $\alpha \omega N/(qN^2)$. The cutoff modulus per coordinate time is

$$E_{\Lambda} = \frac{1}{2} \sum_{\kappa \geq 1} 2\kappa \int_{\mathbb{R}} \frac{d\omega}{2\pi} \text{Tr}_{x,4} E_1(D_{\omega}^2/\Lambda^2).$$

Frequency integration reproduces the known cylinder vacuum response. Independent metric variations conserve the static stress and recover the established Dirac heat coefficients $a_0=4$, $a_2=-R_E/3$ through a_4 [29]. Those coefficients are not a Nested-Space discovery. The specified ultraviolet subtraction is

$$E_{\text{sub}}(M) = E_{\Lambda} - \frac{A_0\Lambda^4/2 + A_2\Lambda^2 + A_4\log(\Lambda^2/M^2)}{32\pi^2}.$$

At $a=1$, $R=2$ and $M=1$, the remainder energy is about -0.09050 with neck null stress about -0.00911 at $\Lambda=4$. This is a calculated negative null contribution under that subtraction, not a scheme-independent complete source ([covariant source](#)). On the same lapse/radius/shape probes the determinant now supplies actual projections $b_A^{\text{determinant}}$ in

$$b_A^{\text{determinant}} + \int A_i C_i + b_A^{\text{compensator, link, state}} = 0.$$

No coefficient is fitted to hold the neck open. Uniform clock scaling multiplies the covariant energy, while a fixed spatial-Hamiltonian cutoff does not.

Repository derivation: normalization profiles. Write $x=d^2/\Lambda^2$. The calculated modulus per eigenvalue is half the exponential integral $E_1(x)$. A smooth weighted logarithm and the heat-covariantized old rank term obey

$$\begin{aligned} g_{\log} &= -\frac{1}{2}e^{-x}\log(d^2/M^2) \\ g_{\log} &= g_{\text{PT}} + [\log(M/\Lambda) + \gamma_E/2]e^{-x} + h(x) \\ h(x) &= -\frac{1}{2}[e^{-x}(\log x + \gamma_E) + E_1(x)]. \end{aligned}$$

The remainder is not a multiple of the identity or of the heat kernel, so replacing the old finite-rank term by a heat trace changes the finite action. The identity is a comparison of prescriptions, not an insertion into Γ_{one} ([normalization](#)). Andrianov's primary projector $P_N=\Theta(1-D^2/\Lambda^2)$ is a different object [11]; its cylinder cutoff wall has a closed form, while an unrestricted $\text{Tr}1$ integral diverges.

Repository derivation. Assign half of the cross-region link energy to each region, with $h_A=(P_{AH}+HP_A)/2$, and use $C_{(ij)}=(c_j^{\dagger}c_i)$. Then

$$\frac{dE_A}{dt} = \text{Tr}(C i[H, h_A]) + \text{Tr}(C \dot{h}_A).$$

For the static vacuum, $[H, C]=0$, so continuing net regional injection is zero. A prepared excitation transports conserved energy through the same geometry-derived link ([energy transfer](#)). Its current depends on occupations and correlations as well as the retarded map.

A prescribed smooth radius pulse gives exactly $H(t)=H_0+\epsilon g(t)V$. It creates Dirac particle-antiparticle excitations. Independent evolution and spectral transition calculations agree, and supplied geometric work accounts for the excitation energy ([vacuum work](#)). At $\epsilon=0.02$ and $\tau=0.5$, one representative $\kappa=1$ channel produces approximately $1.40181e-5$ pairs and energy $3.46640e-5$ in throat units. These are one-channel finite-amplitude controls, not cosmological rates. A uniform clock rescaling produces no pairs.

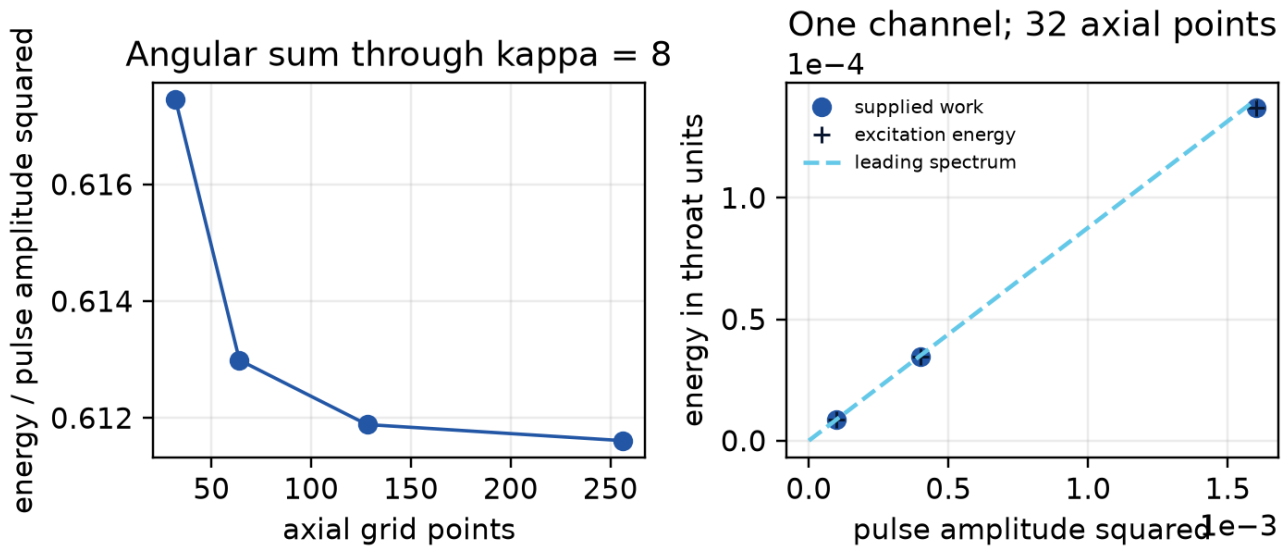


Figure. Response to a prescribed geometry pulse. Left: spatial refinement of the angular sum. Right: independent finite evolution accounts for supplied work in one channel. The pulse is an input; this is not a self-sourced cosmology. [Vacuum-work evidence](#).

This is a calculated conversion from supplied geometric work into Dirac excitations. The deformation's own source and backreaction remain to be solved. A null-sign match, pair creation or a conserved finite energy ledger alone is not the complete metric equation.

Repository derivation: normalized histories and geometric noise. The same covariant operator supplies the canonical Hamiltonian $H=N^{(1/2)}H_0 N^{(1/2)}$. For a Gaussian fermionic covariance $0 \leq C \leq I$ and two history unitaries, the normalized closed-time-path amplitude is

$$Z[+, -; C] = \det(I - C + CU_-^\dagger U_+)$$

$$\Gamma_{\text{IF}} = -i \log Z.$$

The Fock-space trace identity is due to Klich [30]; the present record implements it for this geometry vertex and checks mixed and pure states against an explicit eight-dimensional Fock calculation. Equal histories give $Z=1$. Causal contour derivatives reproduce the retarded commutator response, and the symmetrized noise matrix is positive. For the Gaussian time window already used in the pulse calculation, the smeared vertex variance equals the leading pair-production coefficient:

$$\text{Var}(V_f) = N_{\text{pairs}}^{(2)} / \epsilon^2.$$

At $\tau=0.5$ this common weight is 0.03533459211 on the refined spatial grids ([influence](#)). Bare mean forces and equal-time fluctuations do not converge in the same way; they remain finite-regulator quantities, not renormalized absolute stress. Multiplying by a real local-action phase preserves equal-history normalization while changing the mean source, so $Z=1$ does not select the physical finite terms. The influence approach to gravitational response has prior work by Martín and Verdaguer [31]. The contribution here is the executable connection to this smooth Dirac geometry and its previous energy/pair calculation.

Repository derivation: matching static and causal response. The radius channel now connects the Euclidean frequency operator to the Lorentzian Hamiltonian response. Independent matrix integration of the ordinary finite Dirac loop equals the spectral vacuum susceptibility at imaginary

external frequency, to about $10^{(-15)}$ on the tested finite grids. The finite proper-time Hessian is calculated separately, with its regulator retained.

The periodic-minus-antiperiodic difference removes state-independent local metric terms on identical geometry. It permits the continuum matching control

$$K_E^{P-AP}(\nu) \longrightarrow \chi_R^P(i\nu) - \chi_R^{AP}(i\nu).$$

The independent retarded sum uses a positive Abel transition weight and a stated cutoff extrapolation. These finite regulators are distinct. Through angular label $\kappa=8$, the two methods give:

Imaginary frequency	Euclidean relative radius response	Retarded extrapolation minus Euclidean
0	0.185771550	-1.49e-9
0.6	0.172525449	-1.41e-9
1.4	0.130625412	-1.14e-9

Spatial, angular, internal-frequency and cutoff refinements are recorded separately ([response matching](#)). The omitted angular tower has no proved remainder bound. A change from inverse-radius to log-radius coordinates adds the required contact (H''); its relative value is nonzero and is retained. With that term, independent metric differentiation agrees with the earlier covariant source Hessian. This is a verified matter-response connection in the smooth ultrastatic radius channel. Absolute finite coefficients, finite-cutoff causal completion and constrained metric backreaction remain to be determined.

5. Recursion and cosmological predictions

At a common dimensional energy, parent-normalized first-order elimination gives

$$\Gamma_p(x) = K_p(x) - \frac{1}{\Omega} b \Gamma_c(x/\Omega)^{-1} b^\dagger.$$

Here $b=B_{(dim)}/\Lambda_p$. The inheritance scale $\zeta=(\Lambda L_*)^2$ and cutoff ratio $\Omega=\Lambda_c/\Lambda_p$ remain distinct. Neither is selected by unit dilation alone. Finite chains and spatial endpoint estimates are verified; the covariant state, physical recursive domain and stationary scale remain open.

Let Q_b denote internal transfer into the visible component and J_b, J_d denote external room supply. The complete ledger is

$$\begin{aligned} \dot{\rho}_b + 3H(\rho_b + p_b) &= Q_b + J_b \\ \dot{\rho}_d + 3H(\rho_d + p_d) &= -Q_b + J_d \end{aligned}$$

$$\dot{\rho} + 3H(\rho + p) = J_b + J_d.$$

Internal Q_b cancels. The Bianchi identity requires the full source of ordinary Einstein gravity to be conserved; external room supply therefore needs its reservoir/boundary gravitational accounting. A throat surface integral is power, and a worldtube integral is energy. A cosmological density rate additionally requires proper time, proper volume and deposition into the identified component.

For the closed flat constant-G baseline with visible dust, a constant fraction f below unity requires

$$\frac{Q_b}{H\rho_b} = 1 - 2q_{\text{dec}}.$$

Zero Q_b alone does not force deceleration: dust plus positive Λ has zero exchange and approaches de Sitter expansion. A finite regular plateau is an additional condition. A known scalar Q_b also leaves pressure and perturbations undetermined ([plateau conditions](#), [source requirements](#)).

Predicting $H(z)$, BAO and CMB requires the background and thermal history. Predicting clustering and lensing also requires pressure perturbations, anisotropic stress, momentum transfer and constrained metric response. Λ -CDM is a predictive comparison model; CPL is a phenomenological equation-of-state parameterization. Neither is replaced merely by naming an unevaluated Q .

Current comparison targets are documented with their model assumptions in [observational targets](#). H0DN reports 73.50 ± 0.81 km/s/Mpc, with significance depending on the comparison [23]. DES Y6 reports $S_8 = 0.789 \pm 0.012$, a 2.6-sigma projected difference from its primary-CMB comparison [24]. KiDS-Legacy reports $S_8 = 0.815$ with $+0.016/-0.021$ uncertainty, agreeing with Planck at 0.73 sigma [25]. These are development targets, not NSC predictions or evidence that a boundary mechanism has resolved a tension.

6. Nuclear physics and the remaining common-action equations

The Skyrme/BPS sector is a possible low-energy construction pattern. Its coefficients must be obtained from the evolved common effective action before its particles, binding energies or stellar limits are predictions of NSC.

The often-quoted 3.34 solar-mass maximum is imported from a nuclear-calibrated Einstein-BPS model [5]. The historical runner records the published value rather than independently recomputing it. Its potential and calibrated nuclear coefficients matter; other potentials yield different maxima. It is not a universal collapse threshold or an unfitted NSC result ([stellar benchmark](#)).

The next coupled calculation must determine: the full spinor boundary interaction and its domain; the remaining finite coefficients, compensator, physical state and link of the specified ultraviolet-subtracted determinant source; the independent metric equations; and the physical response around that solution. The present source is that specified determinant remainder, not a scheme-independent completed stress. Only then can particle identities, nuclear interactions, collapse, and cosmological predictions be established with the same parameters.

The joint closure is a candidate project contribution, not an established novelty claim. Primary ingredients retain their original attribution. Failure of one local realization informs its revision; it neither disproves the nested architecture nor confirms it.

7. Reproduction and scope

This working revision contains 81 compact records: 58 frozen historical records and 23 scoped follow-ups. All new scientific records, their generators and hash-bound inputs are imported from committed laboratory evidence. Each has a declared comparison policy; source and scope fields remain exact. Schema adapters live outside immutable scientific generators. The source and normalization/state records compare every field with relative tolerance 2×10^{-8} and absolute tolerance 2×10^{-7} ; the response-matching record uses 2×10^{-8} for both tolerances.

```
make demonstrate
make verify
make reproduce-exact
```

The short demonstration follows computed boundary response, vacuum work, and the invariant spinor sector. The full gate authenticates the graph, runs focused tests, reproduces all records in isolation, and verifies two identical PDF builds. Same-environment exact reproduction and portable numerical comparison are separate checks. See [reproducing](#) and the [result manifest](#).

The following appendices preserve the main normalization corrections and finite mathematical controls. Their historical numerical roots remain diagnostics. Full covariant physical closure, predicted particle species, a measured dark fraction, and our universe's parent/child identification remain open.

Appendix A. Historical scale tests and corrections

The original numerical candidates are retained for provenance. This appendix does not define the current complete action or imply that adding a recursive tail alone supplies the missing finite measure and state.

A.1 Historical scale candidates

The inheritance scale is

$$\zeta = (\Lambda L_*)^2$$

$$\mu^2 = 1 - \frac{3\pi}{2\zeta}.$$

Repository derivation. Matching the local on-room vacuum-form component *108/1015* to the doubled gap, then using the exact expanding child's Ricci sign, selected a *conditional* one-scale gap $|\Phi|^2/\Lambda^2=1006/1015$ and a boundary cross-projection $\backslash Xi_{(boundary)}=54/503$ ([nsc-1-s-one-child-orientation.json](#)). **Repository derivation.** The exact child asymptotic rejects that magnitude. The actual child vacuum is $\Lambda_{(e,child)}L_*^2=18\pi$, so a real positive gap requires $\zeta>3\pi/2$. Direct $\Lambda L_*=1$ gives a negative μ^2 and is not the physical child-matched branch ([nsc-1-s-one-child-scale-correction.json](#)). The identities for Φ as gap, Schur self-energy, and metric link remain. The numbers *1006/1015* and *54/503* are **superseded** as child-matched predictions.

The ZETA1 chain then searched for a child-allowed stationary scale of the joined-minus-disconnected Dirac spectrum on the global slice. Every root in the following table is a **numerical diagnostic**. None is a physical ζ . The later anomaly compensation supersedes all of them as stationary points of the one equation.

Compact record	What was computed	Diagnostic value	Why it is not physical ζ
nsc-2-zeta1-lowest-mode.json	Isolated lowest-mode heat minimum	$\zeta=5.096657218448003$, $\mu=0.27458356021684416$	Lowest sector only; heat trace, not the determinant; superseded candidate .
nsc-2-zeta1-angular-tower.json	Angular tower through $n=12$	Diagnostic $n=1$ heat root 5.096622100098284 ; $n=2$ removes that heat-only root; $n \leq 12$ minimum derivative 7.350807275212887	Heat-only truncation; superseded candidate ; absence of a heat root is not a theorem against stationarity.
nsc-2-zeta1-regulated-determinant.json	Exponentially regulated joined-minus-disconnected determinant	$\zeta_{(\backslash det)}=4.99072409354754$, $\mu_{(\backslash det)}=0.2361577587050819$	Uncompensated determinant; warp, lapse, shift, and anomaly omitted; superseded candidate .
nsc-2-zeta1-y-boundary-sensitivity.json	Factorized compact-y controls	Periodic root 4.8765555133228515 ; Neumann root 4.79786026064662 ; Dirichlet and MIT bag: no root	Factorized unwarped control; a zero mode was not selected in order to force a root; superseded candidate .
nsc-2-zeta1-orbifold-parity.json	Joint fermion/metric $S^{1/2}/Z_2$ parity	$\zeta_{(orb)}=4.747863009733466$, $\mu_{(orb)}=0.08643829078229333$	Unwarped factorization; superseded candidate .
nsc-2-zeta1-warped-y.json	Exact conformal compact warp	$\zeta=4.748389947082489$, $\mu=0.08707307719677547$	Factorized radial-plus-y sum, not the full Dirac operator with lapse and shift; superseded candidate .

Every corresponding JSON record states that the candidate is not promoted, that lapse and shift or the local matrix anomaly remain omitted, and that the electron mass is not predicted.

A.2 Anomaly and inherited subtraction

Repository derivation. The invariant partition function requires cancellation of pure cutoff rescaling while retaining the physical change of the child-constrained gap. For one eigenvalue the physical child-link derivative after compensation is negative whenever the gap constraint $\mu^2=1-3\pi/(2\zeta)$ is imposed ([nsc-2-zeta1-anomaly-decomposition.json](#)).

At the warped determinant candidate $\zeta=4.748389947082489$:

Piece	Value
Determinant derivative	0.01201522080638507
Pure cutoff anomaly	12.336590431250533
Physical child-link derivative	-12.324575210444092

The compensated child-link derivative stays negative on the 257-point scan $3\pi/2 < \zeta \leq 200$, from -19.422735759304963 to -0.5875504421475721.

Those historical numbers used the inconsistent conversion $(\lambda_j + \mu^2)/\zeta$. They remain numerical diagnostics of that proxy.

Therefore those determinant-only ζ values are invalidated as stationary points of the one equation. $\zeta_{(det)}$ and $\zeta_{(warped\ y)}$ are uncompensated determinant diagnostics, not inheritance-scale predictions.

Historical interpretation. The next missing contribution was identified as the recursive child tail, rather than an independently weighted Einstein-Gauss-Bonnet or heat action ([nsc-2-zeta1-anomaly-owner-correction.json](#)). Geometry is already the compensating anomaly of the same fermionic determinant. Adding that geometry again is forbidden double-counting. The calculations through the scan joined one parent to one child treated as a terminal asymptotic domain. Recursion requires that the child's boundary response already contain its own child tail. The compact nonclaims include `recursive_tail_is_known_to_restore_a_scale_root: false`.

Related routes already rejected on their own JSON nonclaims, and not repaired by retuning Φ or adding a second geometric weight:

- $\Lambda L_* = 1$ as the physical child-matched gap;
- the lowest-mode heat-only scale minimum;
- independently weighted Einstein-Gauss-Bonnet or heat geometry as the missing scale owner;
- pure critical five-dimensional Einstein-Gauss-Bonnet as the complete interface law;
- the standalone bosonic heat-trace graviton covariance as a positive Stieltjes propagator ($g''(0)=228/175>0$; [nsc-1-s-one-reflection-positivity-obstruction.json](#));
- the inverse relative heat Hessian as Osterwalder-Schrader positive metric covariance (the full 31-time reflection matrix has minimum eigenvalue -1.3679466754324912; the source record counts fourteen negative modes while portable Linux counts thirteen because one near-zero mode crosses the numerical counting tolerance; [nsc-1-s-one-relative-reflection-test.json](#));
- constant two-wall shadow matter as the dominant kiloparsec gravitating mass.

The fermionic sheet-resolution observable is Osterwalder-Schrader positive in the controlled flat sector ([nsc-1-s-one-fermionic-relative-observable.json](#)). That does not make σ_3 the complete graviton, and it does not restore a scale root. Full-exponential positivity in that sector is a heat-kernel diagnostic, not the complete physical anomaly.

A.3 Corrected units and the sign theorem

Repository derivation. The ZETA1 definitions are $\lambda_j = L_*^{-2} \text{eig}_j(D_{\text{spatial}})^2$, $\zeta = \Lambda^2 L_*^{-2}$, and $\mu^2 = |\Phi|^2 / \Lambda^2 = 1 - a/\zeta$ with $a = 3\pi/2$. The same declared additive-gap operator therefore requires

$$q_j = \frac{\lambda_j}{\zeta} + \mu^2 = 1 + \frac{\lambda_j - a}{\zeta}.$$

The historical runners implemented $(\lambda_j + \mu^2)/\zeta$, a physical mass squared smaller by $1/\zeta$ than the declared mass squared. Compact record [nsc-2-zeta1-unit-closure-check.json](#) reconstructs the old derivative as a control, then changes only that conversion. It retains radius 30, 750 radial half-intervals, angular sectors 1-12, 32 compact intervals, and the historical factorization and angular weights.

At the old warped candidate $\zeta = 4.748389947082489$:

Quantity	Historical argument	Argument with consistent units
Determinant logarithmic derivative	0.01201522080638507	-24.319952223294422
Derivative after inherited cutoff subtraction	-12.324575210444092	-36.58292832809548

The independent old-argument reconstruction differs from the stored derivative by 1.53×10^{-8} . With radial spacings 0.08, 0.04, and 0.02, the corrected subtracted derivative is -36.58813190, -36.58292833, and -36.58162543. Its sign is resolved.

Repository derivation. For each mode the determinant piece is one half of $E_1(q_j)$. At fixed spatial geometry, subtracting the same pure-cutoff term used historically gives

$$\frac{d\Gamma_{\text{sub},j}}{d\log \zeta} = -\frac{e^{-q_j}}{2q_j}.$$

The disconnected radial matrix is the principal submatrix obtained by deleting the throat node. Hermitian eigenvalue interlacing and the positive decreasing weight $f(\lambda) = e^{-q(\lambda)}/q(\lambda)$ then imply that the total subtracted derivative is strictly negative for every $\zeta > 3\pi/2$ in this finite, frozen-geometry, additive-gap family. The 257-point scan corroborates the sign; it is not the basis for extending the statement between sample points. A diagnostic root of the unsubtracted finite determinant near $\zeta = 6.09675392014$ still has subtracted derivative about -14.15469, so it does not solve the adopted subtracted scale equation. No physical scale is inferred.

Consequence. Refining this family or selecting another determinant-only minimum cannot close its scale equation. A further attempt needs a derived change in the physical operator, its geometry/link dependence on scale, or the anomaly prescription. Merely naming the omitted recursive tail does not prove that it supplies that change.

Repository derivation. For the same relative-sheet symbol

$$D_\delta = \begin{bmatrix} p e^{(-\delta)} & \Phi \\ \Phi & -p e^{(\delta)} \end{bmatrix}$$

the unregulated finite determinant is independent of δ . The proper-time functional used later in ZETA1 has a strictly positive regulated relative Hessian at $\delta=0$ for $p \neq 0$. Raw determinant invariance therefore cannot justify setting the regulated relative Hessian to zero. The primary anomaly paper distinguishes its normalization scale from the spectral cutoff ([arXiv:1106.3263](#)).

Appendix B. Finite operator and boundary controls

B.1 Finite regulated variations and a spacetime obstruction

Repository derivation. For a Hermitian finite Dirac matrix with no unresolved zeros, one explicit proper-time prescription is ([nsc-3-regulated-recursion.json](#); [docs/nsc-regulated-recursion.md](#))

$$\Gamma_{\Lambda, M}(D) = \frac{1}{2} \text{Tr} E_1(D^2/\Lambda^2) + N[\log(M/\Lambda) + \gamma_E/2].$$

Its first and second variations retain noncommuting perturbations through divided differences. A finite scale cocycle $C(t) = \Gamma_{\text{reg}}(D_t) - \Gamma_{\text{reg}}(D) - t \text{Tr} \text{Sigma}$ is derived rather than discarded; its endpoint matches an independent integral of $C'(t)$. Normalization M and cutoff Λ remain distinct. This finite orthonormal-basis calculus is not a derived continuum gravitational measure.

Repository derivation. An ultrastatic Euclidean product $D_E^2 = -\partial_\tau^2 + H^2$ produces a frequency factor $\text{Tr}(sp)e^{(-tH^2)/\sqrt{4\pi}t}$. That factor changes the variational functional; a spatial heat trace cannot silently inherit a four-spacetime anomaly.

Imported result / repository witness. The recorded horizon-penetrating coordinate-time Hamiltonian is not elliptic in the trapped region: at the throat $\beta^2 = 3\pi/2 > 1$, and a nonzero spatial covector makes one principal eigenvalue exactly zero. Elliptic spatial heat calculus therefore cannot be imported as the full spacetime determinant. The general loss of ellipticity at horizons is documented by Finster and R  ken ([arXiv:1512.00761](#)); the project contribution is the explicit witness for this carrier, not discovery of the phenomenon.

Along $D(t) = e^{(-t)}D$ at fixed cutoff, normalization, and finite domain, $d\Gamma/dt = \text{Tr} e^{(-D(t)^2/\Lambda^2)} > 0$. No isolated common-scale extremum exists in that finite family. Normalized finite recursion with $1/\Omega$ agrees with direct chain inversion in declared-matrix controls. H and b there are not yet computed throat maps. The compact nonclaims include `full_covariant_action_derived: false` and `physical_stationarity_solved: false`.

B.2 The isolated radial throat is gapless

Repository derivation. The unwarped spatial radial Dirac operator already declared by the project,

$$D = -i\sigma_2\partial_\rho + \sigma_1\frac{\kappa}{\sqrt{1+\rho^2}}$$

has essential spectrum equal to R ([nsc-3-radial-spectrum.json](#); [docs/nsc-radial-spectrum.md](#)). A constructive Weyl sequence of compactly supported H^1 bumps escaping to infinity shows that every real energy belongs to the essential spectrum. Smooth radial geometry and current-conserving gluing alone do not generate a nonzero spectral mass gap. Finite-box levels are not physical masses. There is no global L^2 zero mode. The result does not cover the compact warped, interacting, recursive, or Lorentzian operator, and it does not reject Nested-Space Cosmology.

B.3 A controlled geometric gap from repeated throat geometry

Repository derivation. Repeating the finite segment $\rho \in [-R, R]$ with $w = (1 + \rho^2)^{-1/2}$, instead of two asymptotically widening ends, produces an open spectral gap without an inserted constant mass ([nsc-3-geometric-chain.json](#); [docs/nsc-geometric-chain.md](#)). At zero energy the exact motif transfer multipliers are $\exp(\pm 2asinh R)$; neither has modulus one, so zero is absent from every real Bloch fibre. Continuum first band edges and second-order lattice agreement are

R	Continuum first band edge	Finest-lattice absolute error
2	0.7034881641	6.88×10^{-6}
4	0.4552655377	6.20×10^{-6}
8	0.2479700183	2.43×10^{-6}

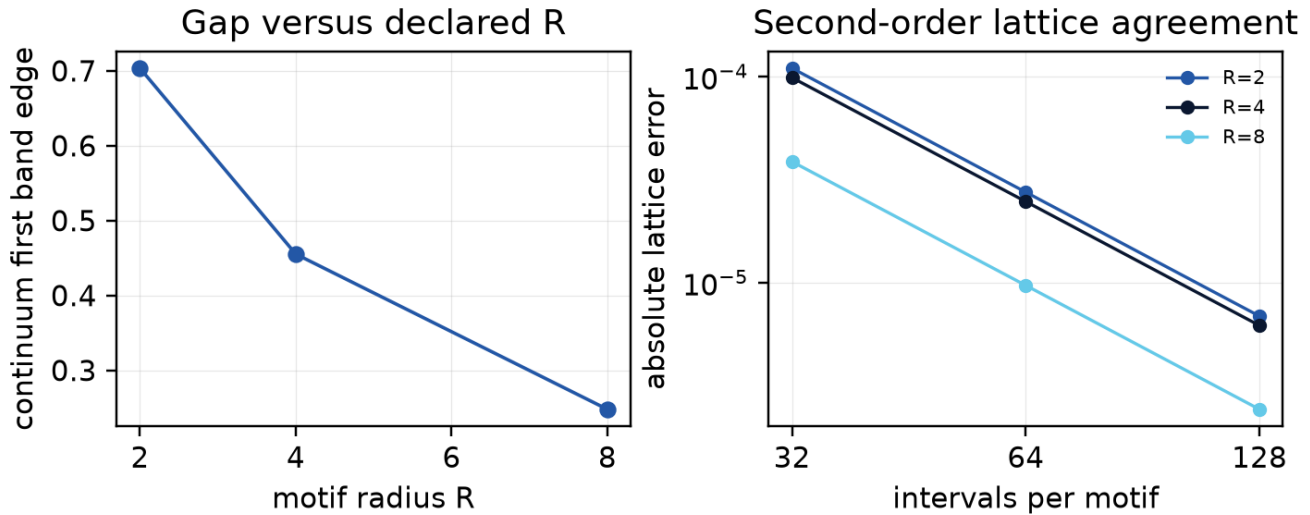


Figure. Continuum first band edges and lattice errors from [nsc-3-geometric-chain.json](#). R is a declared input.

The next-room link is the staggered stencil entry $B_{(last\ edge, first\ node)} = 1/h + w_{(last)}/2$, not a fitted mass. Normalized finite recursion with that geometric B agrees with direct multi-motif inversion. R and Ω remain declared inputs. Seams are periodic in w but gravitational junction equations are unproved. This is a positive project-level controlled realization of geometric confinement, not a new-to-world Floquet discovery and not a Lorentzian nested cosmology or an electron.

B.4 Computed finite boundary maps and the threshold distinction

Repository derivation. The finite first-order radial calculation now evaluates parent and child Weyl maps with sparse LU, throat reconstruction, independent ODE integration, and direct Schur controls ([nsc-3-boundary-response.json](#)). The parent lies at positive radial coordinate and its outward throat normal is negative; the child lies at negative radial coordinate and its outward throat normal is positive. The oriented relation is

$$N_p(E) + N_c(E) = E[m_c(E) - m_p(E)].$$

A nonzero real-energy probability-current probe tests the local conservation law. The compact warp calculation reports mode-mixing leakage: its average-warp expression is a projected approximation. These results compute finite spatial maps; the full horizon boundary maps, physical compensator, and Lorentzian geometry equations remain open.

Repository derivation. The same isolated radial geometry supplies an explicit threshold counterexample. Solving the upper squared-Dirac equation with unit throat value and vanishing exterior value gives, for $t = \text{asinh}(R)$,

$$I_p = e^{3t}/6 + e^t/2 - 2/3$$

$$I_c = 2/3 - e^{-t}/2 - e^{-3t}/6$$

$$N_p + N_c = I_p^{-1} + I_c^{-1} \rightarrow 3/2.$$

Independent ODE solutions reproduce these maps ([nsc-3-threshold-response.json](#)). The positive boundary stiffness coexists with the gapless radial bulk spectrum; it is not a particle mass of 3/2. A mass identification requires the full pole problem, residues and physical state. At zero energy $m(E)$ can be singular, so $E m(E)$ must be evaluated by its limit or by the squared-operator boundary problem.

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